

and are also used for PCB inspection. High-density ICs and BGAs (ball grid arrays) are difficult to test with AOI systems as they have connections underneath the packages. AXI systems use X-rays instead of visible light and thus are able to inspect the elements that are not visible to AOI systems.

High-focus systems with sub-micron spot sizes are ideal for the MEMS, opto-electronics, mobile phones, telecom electronics, BGAs and loaded PCBs. 3D imaging has increased AXI performance manifold, as AXIs are having angled 3D imaging with optional computed tomography (CT) system for reconstruction of 3D images to gain more insight into components and sub-assemblies. Higher magnification for zooming in on specific areas with user-friendly software adds to the AXI testing ranges.

In-circuit testing

In-circuit testing (ICT) is a comprehensive PCB testing technique, which measures circuit parameters such as resistance and capacitance, along with the operation of analogue components such as operational amplifiers and some functionality of digital circuits.

ICT systems gain direct electrical access to the components on a PCB via an electromechanical 'bed of nails' fixture and test for manufacturing faults. Advancements in ICT systems have contributed towards lowering overall manufacturing test costs by improving fault coverage, reliability and throughput of in-circuit production tests.

Fully automated ICT systems having compact size and integration with high-volume production lines minimise the need for operator handling. This saves labour costs and reduces the risk of product damage due to electrostatic discharge (ESD). Capabilities of ICT systems have advanced far beyond when they were first introduced. Now, with all the additional electrical test capabilities that do not require actual physical test access, we can categorise today's versatile test systems as 'electrical test controllers' rather than 'in-circuit testers.'

Vendor	AOI Solution	Key Features
CyberOptics	QX600	Strobed inspection module (SIM) with enhanced illumination using LED lighting Faster and simpler programming with AI ² (autonomous image interpretation) technology
Omron	VT-S500	Repair station functionality Data analysis and quality control software tools Q-upNavi-integrated software tools for the monitoring and improvement of each stage of the manufacturing process including solder paste screen printing, placement, reflow, post-solder inspection and X-ray.
Mirtec	MV9 2D-3D	Exclusive 15 megapixel ISIS vision Scalable resolution from 20 microns per pixel to 10 microns per pixel 3D multi-frequency Quad Moiré technology for true 3D inspection of SMT devices on finished PCB assemblies
Mek	PowerSpector	Can be equipped with nine cameras (one top and eight side) Most versatile inspection engine Multiple light sources from multiple angles
Saki Corp.	BF-3DI	Unique optical design and four-way projection enables realisation of all parts 3D inspection with superb detection ability

Vendor	AXI Solution	Key Features
Omron	VT-X700	Selectable resolutions of 10, 15, 20, 25 or 30 microns combined with selectable numbers of X-ray projections high precision, X-ray, Angled 3D CT imaging
Saki Corp.	BF-X2	Automated high-resolution 3D measuring technology Planar CT technology to get high-precision 3D data
Geopel Electronics	OptiCon X-Line 3D	Optimal adaptation to future components and manufacturing requirements via Software X-PILOT GigaPixel technology for image capture
NordsonYestech	X3-3D X-Ray	Digital tomosynthesis technology for capturing multiple images in different slice heights
Matrix Technologies	X2	Slice filter Technology (SFT) for double-sided PCB assembly and component detection Three axes programmable motion system with variable field of view-size. Barcode scanner (1D/2D) for serial number and product-type selection

"Although ICT is appreciated for component-level testing, testing of components that are connected in parallel is almost impossible," adds T. Anand, managing director, Knewron.

ICT to ETC transition. ICTs have been popular since long due to their usage in high-volume production environments, mostly for simple tests. "However, ICT is the most tedious, cumbersome and expensive type of testing. Creating an ICT fixture is costly and time-consuming activity. It is typical example of White Box (Glass Box) testing. Now, electrical test

controllers (ETCs) aka automated test controllers ATCs do much better job than ICTs. Their testing method can be categorised as Black Box (full function) testing too," says T. Anand.

ETCs do excellent job in simulating end-user, which gives more valuable information than ICTs. In short, ICTs have grown up to be ETCs.

Typical digital ICTs deploy boundary scan technique and perform basic digital verification, unlike ETCs where test case is generated, voltage/data is applied to respective pins and output is read and verified.